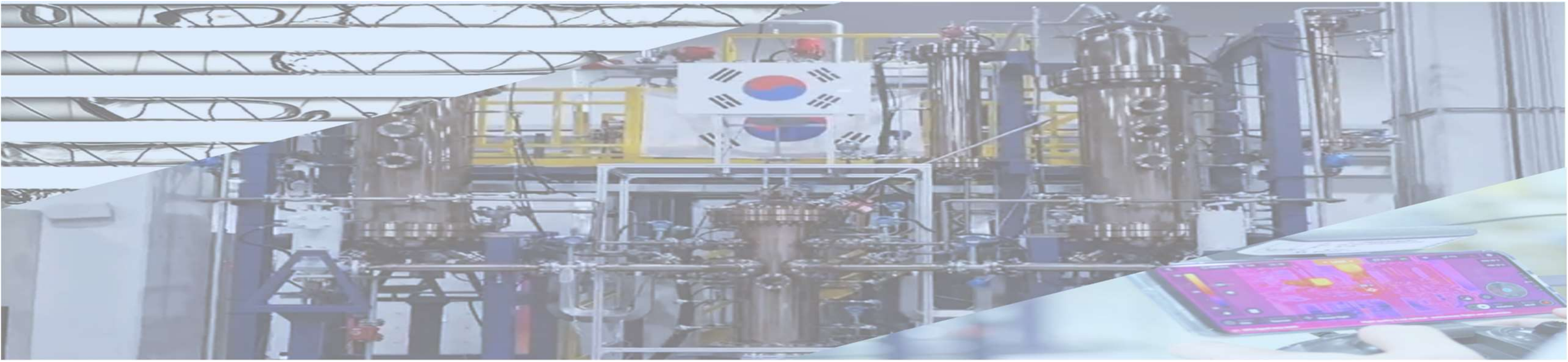

Steady State Analysis of two MSRE Modules with Molten Salt Properties application to MARS-KS

2026. 07.

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Nuclear **T**hermal-**H**ydraulics and **R**eactor **S**afety Laboratory (**NTHRS Lab.**)

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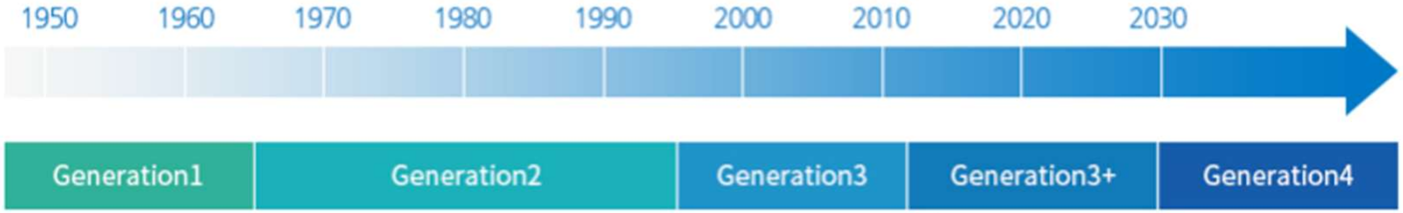
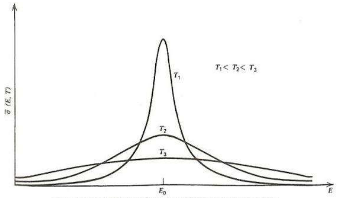
1 Introduction

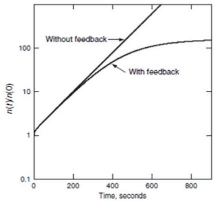
Research background



1 Inherent Safety

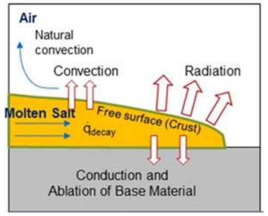


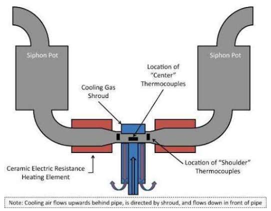
Doppler-broadening of a resonance with increasing temperature.

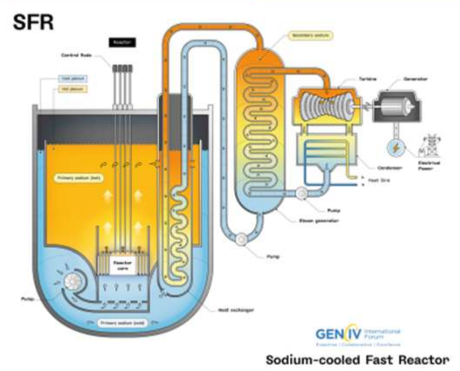


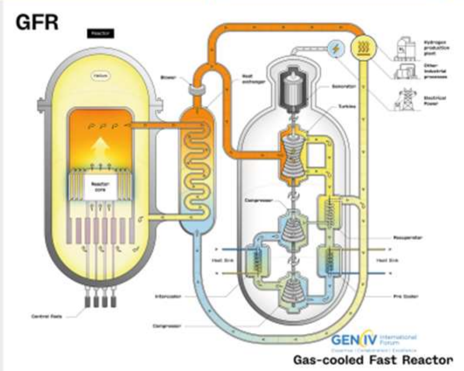
2 High Temperature Operation

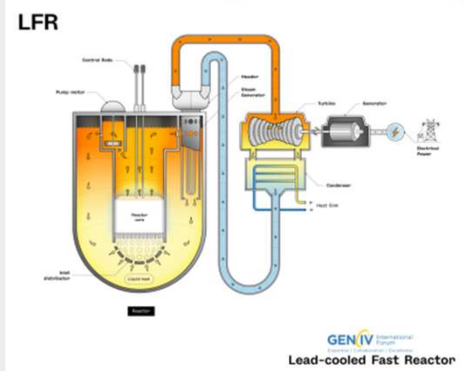
3 Enhanced Safety

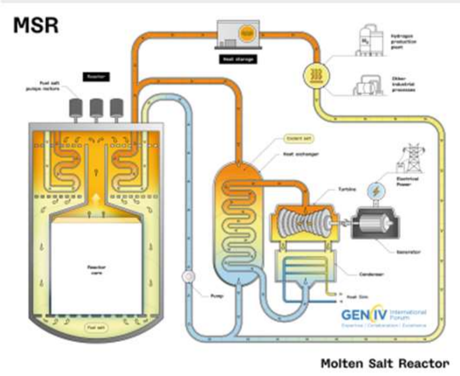


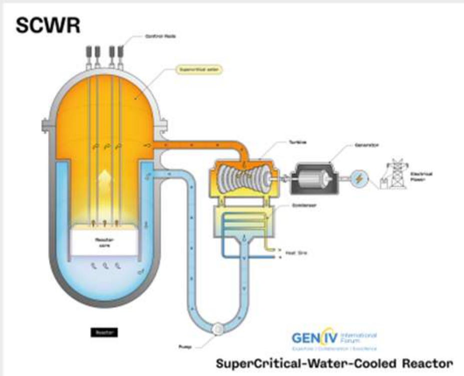


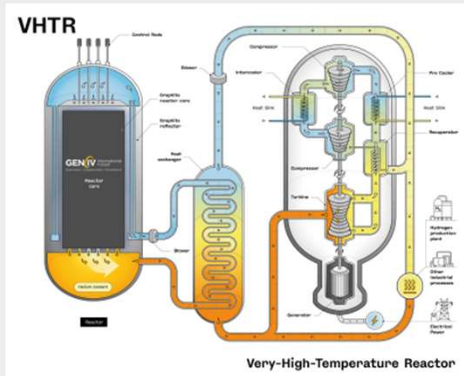






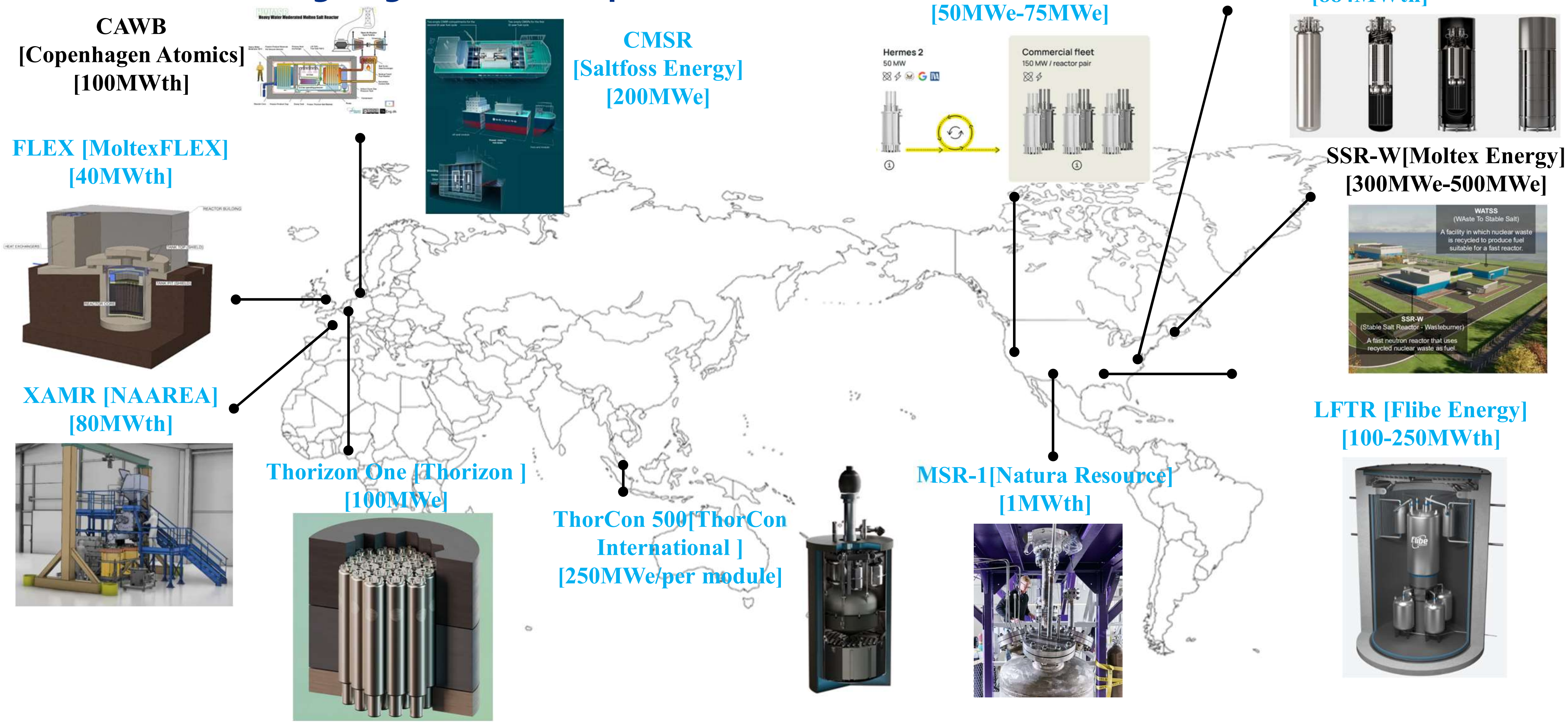






1 Introduction

Ongoing MSR development



Necessity of verifying feasibility of MSR module analysis

- Although efforts have been made to analyze MSRs, the capability to assess the module impact has not yet been demonstrated.

Table. Summary of previous researches to analyze MSR

Institute	Code	Analysis target	Type of fluid	Remark	Module analysis capability
UNIST [2016]	MARS-KS	Natural Circulation Loop [DOWTHERM oil]	DOWTHERM oil	Fluid Property Table Implementation	Not validated
Chinese Academy of Science [2016]	RELAP5	MSRE Core MSBR 1st & 2nd Loop	FLiBe	Point Kinetics Model Implementation	Not validated
Pennsylvania State University & ANL [2019]	NEAMS-SAM	MSRE 1st Loop	FLiBe	Feasibility of SAM Code MSR analysis	Not validated
Poland NCBJ [2021]	TRACE	MSRE 1st & 2nd Loop	FLiBe	Feasibility of TRACE-Serpent analysis	Not validated
Tokyo Institute of Technology [2022]	RELAP5-3D	MSBR 1st & 2nd & 3rd Loop	FLiBe	Analysis of the reactivity insertion condition	Not validated
KAERI [2022]	GAMMA+	MSBR 1st Loop Natural Circulation	FLiBe	Feasibility of GAMMA+ code MSR analysis	Not validated
Terrapower & INL[2025]	RELAP5-3D	Xi'an Jiao Tong University Experiment	60wt% $NaNO_3$ 40wt% KNO_3	Comparison of Heat Transfer Coefficient Calculation Methods	Not validated
Pusan National University [2026]	MARS-KS	MSRE 1st & 2nd Loop	LiF-BeF2-ZrF4-UF4, LiF-BeF2, KCl-UCl3, and FUNaK	Modification of point kinetics model	Not validated

Laboratory Status and Limitations

- Although system-level MSR experimental facilities are under development, they are constructed using water.

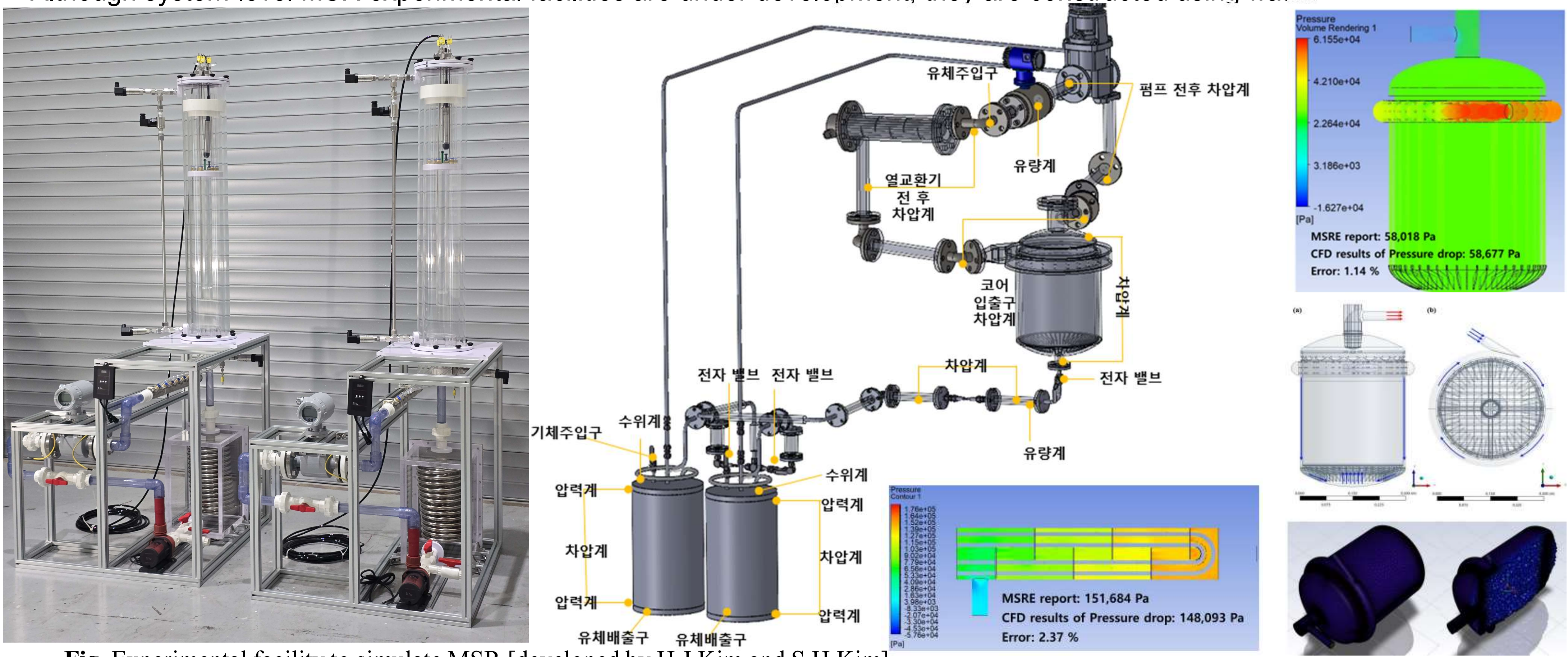


Fig. Experimental facility to simulate MSR [developed by H.J.Kim and S.H.Kim]

1 Introduction

Research objective

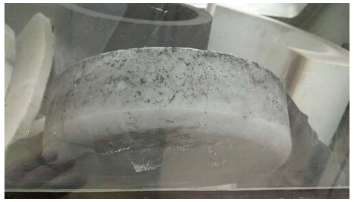
- In this study, the feasibility of MSR module analysis was demonstrated using MARS-KS, compared with the experimental data.

MARS-KS Modification with thermal properties implementation → Comparison with experiment → Multi Units Analysis

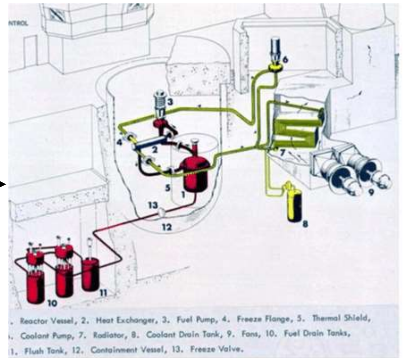
Fluoride fluid

LiF-BeF_2 [FLiBe]

LiF-NaF-KF [FLiNaK]



MSRE



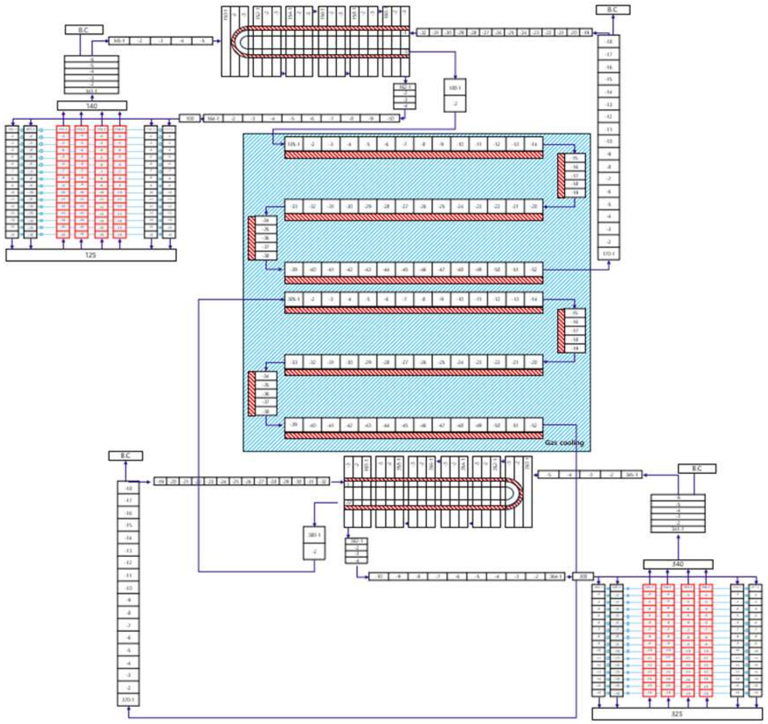
Chloride fluid

NaCl-KCl-$MgCl_2$

NaCl-$MgCl_2$ [Planned]

Drain Experiment [Planned]

Two Modules analysis



Thermal properties implementation method to MARS-KS

MARS-KS Source File Modification to read tpf files

```
DATA wmoles/18.01534d0,20.031d0,44.0098d0,4.0026d0,2.01594d0,     &
            31.9988d0,28.01348d0,22.98977d0,39.948d0,28.9586d0,   &
            210.0d0,                                              &
            18.01534d0,32.3d0,39.4d0,18.01534d0, 80.2d0, 28.9651d0/
DATA fsymbl/'h2o','d2o','co2','he','h2','o2','n2','na','ar',
'dair','lbe','h2on','flibe','flinak','h2onew','nkmg','air2'/
```

```
ELSEIF(nfluid.eq.13) then
   ntg=ntflibe
   npg=npflibe
   nstg=nsflibe
   nspg=ns2flibe
   it3bpg=klpflibe
   it4bpg=klp2flibe
   it5bpg=llpflibe
   nprpntg=nt5flibe
   it3p0g=jplflibe
```

Investigate Temperature & Pressure Dependent Correlation

Engineering Database of Liquid Salt Thermophysical and Thermochemical Properties
Idaho National Laboratory

&

Engineering molten $MgCl_2$–KCl–NaCl salt for high-temperature thermal energy storage: Review on salt properties and corrosion control strategies

MARS-KS Source File Modification [Thermal Properties]

```
DO m=1,nh
    i=ihld(2,ih)
    th=temp(i)
    visc(i)=0.000116d0*exp(3755.0d0/th)
    ih=ih+iscskp
END DO
GOTO 999
```

```
DO m=1,nh
   i=ihld(2,ih)
  !term=max(v_da(i)%Tf,397.7d0)
   thcon(i)=0.629697d0+0.0005d0*term
   ih=ih+iscskp
END DO
RETURN
```

```
DO m=1,nh
   i=ihld(2,ih)
   !term=max(v_da(i)%TsatP-273.15d0,93.d0)
   term=temp(i)
   sigma(i)=0.29578d0-1.2d-4*term
   ih=ih+iscskp
END DO
```

Thermal Properties File Generation [Applicable to MARS-KS]

```
!=================================================================
!singlephase
!=================================================================
        v(1)=1/(2413.0d0-0.488d0*ta)
         v(2)=2077.0d0*ta/pa

!internal energy data from NIST
         u(1)=2415.78d0*(ta-700.0d0) - (0.488d0*ta)*(pa-1.0d5)/(2413.0d0-0.488d0*ta)**2 - pa*(1/(2413.0d0-0.488d0*ta) - 1/(2413.0d0-0.488d0*700.0d0))
         u(2)=3116.0d0*ta
```

Heat transfer coefficient calculation method modification

Gnielinski Correlation

$$Nu = \frac{\left(\frac{f}{8}\right)(Re-1000)Pr}{1+12.7\left(\frac{f}{8}\right)^{1/2}\left(Pr^{\frac{2}{3}}-1\right)}$$

$$f = [0.79 * \ln(Re) - 1.64]^{-2}$$

Thermal Property equation of FLiBe

MARS-KS Source Code Modification

Surface tension $\sigma = 0.29578 - 1.2 * 10^{-4} * T$

Thermal conductivity $k = 0.6297 - 0.5 * 10^{-3} * T$

Viscosity $\mu = 1.16 * 10^{-4} e^{\frac{3755}{T}}$

Property File Generation [Applicable to MARS-KS]

Specific volume $v = \dfrac{1}{2413 - 0.488 * T}$ **Isothermal compressibility** $\beta = 2.3 * 10^{-12} e^{0.001*T} \rightarrow 2.89 * 10^{-10}$ **Specific heat** $C_p = 2415.78$

Internal energy $$U = 2415.78 * (T - T_{ref}) - \frac{0.488 * T}{(2413 - 0.488 * T)^2}(P - P_{ref}) - P\left(\frac{1}{2413 - 0.488 * T} - \frac{1}{2413 - 0.488 * T_{ref}}\right)$$

$$dU = TdS - PdV$$

Thermal expansion $\alpha = \dfrac{1}{v}\left(\dfrac{\partial V}{\partial T}\right)_T$ $\qquad \alpha = \dfrac{0.488}{2413 - 0.488 * T}$

$$dG = -SdT + VdP \implies -\left(\frac{\partial S}{\partial P}\right)_T = \left(\frac{\partial V}{\partial T}\right)_P$$

Entropy $dS = \left(\dfrac{\partial S}{\partial T}\right)_P dT + \left(\dfrac{\partial S}{\partial P}\right)_T dP$ $\qquad s = \displaystyle\int C_p \frac{dT}{T} - \int \left(\frac{\partial V}{\partial T}\right)_P dP$ $\qquad s = 2415.78 * \ln\left(\dfrac{T}{T_{ref}}\right) - \dfrac{0.488}{(2413 - 0.488 * T)^2}(P - P_{ref})$

Thermal Property equation of FLiNaK

MARS-KS Source Code Modification

Surface tension $\sigma = 0.2726 - 1.014 * 10^{-4} * T$

Thermal conductivity $k = 0.36 - 5.6 * 10^{-4} * T$

Viscosity $\mu = 4.0 * 10^{-5} e^{\frac{4170}{T}}$

Property File Generation [Applicable to MARS-KS]

Specific volume $v = \dfrac{1}{2579.3 - 0.624 * T}$

Isothermal compressibility $\beta = 2.89 * 10^{-1}$

Specific heat $C_p = 976.78 + 1.0634 * T$

Internal energy

$$U = 976.78 * (T - T_{ref}) + 0.5317(T^2 - T_{ref}^2) - \frac{0.624 * T}{(2579.3 - 0.624 * T)^2}(P - P_{ref}) - P\left(\frac{1}{2579.3 - 0.624 * T} - \frac{1}{2579.3 - 0.624 * T_{ref}}\right)$$

$$dU = TdS - PdV$$

Thermal expansion $\alpha = \dfrac{1}{v}\left(\dfrac{\partial V}{\partial T}\right)_T$ $\quad \alpha = \dfrac{0.624}{2579.3 - 0.624 * T}$

$$dG = -SdT + VdP \implies -\left(\frac{\partial S}{\partial P}\right)_T = \left(\frac{\partial V}{\partial T}\right)_P$$

Entropy $s = \displaystyle\int C_p \frac{dT}{T} - \int \left(\frac{\partial V}{\partial T}\right)_P dP$

$$s = 976.78 * \ln\left(\frac{T}{T_{ref}}\right) + 1.0634(T - T_{ref}) - \frac{0.624}{(2579.3 - 0.624 * T)^2}(P - P_{ref})$$

Thermal Property equation of NaCl-KCl-$MgCl_2$ [Secondary side of K-MSR]

MARS-KS Source Code Modification

Surface tension

$$\sigma = 0.1330 - 0.48 * 10^{-4} * T$$

Thermal conductivity

$$k = 0.5822 - 2.6 * 10^{-4} * T$$

Viscosity

$$\mu = 0.706 * 10^{-5} e^{\frac{1204}{T}}$$

Property File Generation [Applicable to MARS-KS]

Specific volume $v = \dfrac{1}{1958.84 - 0.56355 * T}$ **Isothermal compressibility** $\beta = 2.89 * 10^{-1}$ **Specific heat** $C_p = 1301.38 + 0.5 * T$

Internal energy

$$U = 1301.38 * (T - T_{ref}) + 0.25(T^2 - T_{ref}^2) - \frac{0.564 * T}{(1958.84 - 0.564 * T)^2}(P - P_{ref}) - P\left(\frac{1}{1958.84 - 0.564 * T} - \frac{1}{1958.84 - 0.564 * T_{ref}}\right)$$

$$dU = TdS - PdV$$

Thermal expansion $\alpha = \dfrac{1}{v}\left(\dfrac{\partial V}{\partial T}\right)_T$ $\alpha = \dfrac{0.56355}{1958.84 - 0.56355 * T}$

$$dG = -SdT + VdP \implies -\left(\frac{\partial S}{\partial P}\right)_T = \left(\frac{\partial V}{\partial T}\right)_P$$

Entropy $s = \displaystyle\int C_p \frac{dT}{T} - \int \left(\frac{\partial V}{\partial T}\right)_P dP$

$$s = 1301.38 * \ln\left(\frac{T}{T_{ref}}\right) + 0.5(T - T_{ref}) - \frac{0.56355}{(1958.84 - 0.56355 * T)^2}(P - P_{ref})$$

MSRE nodalization for MARS-KS

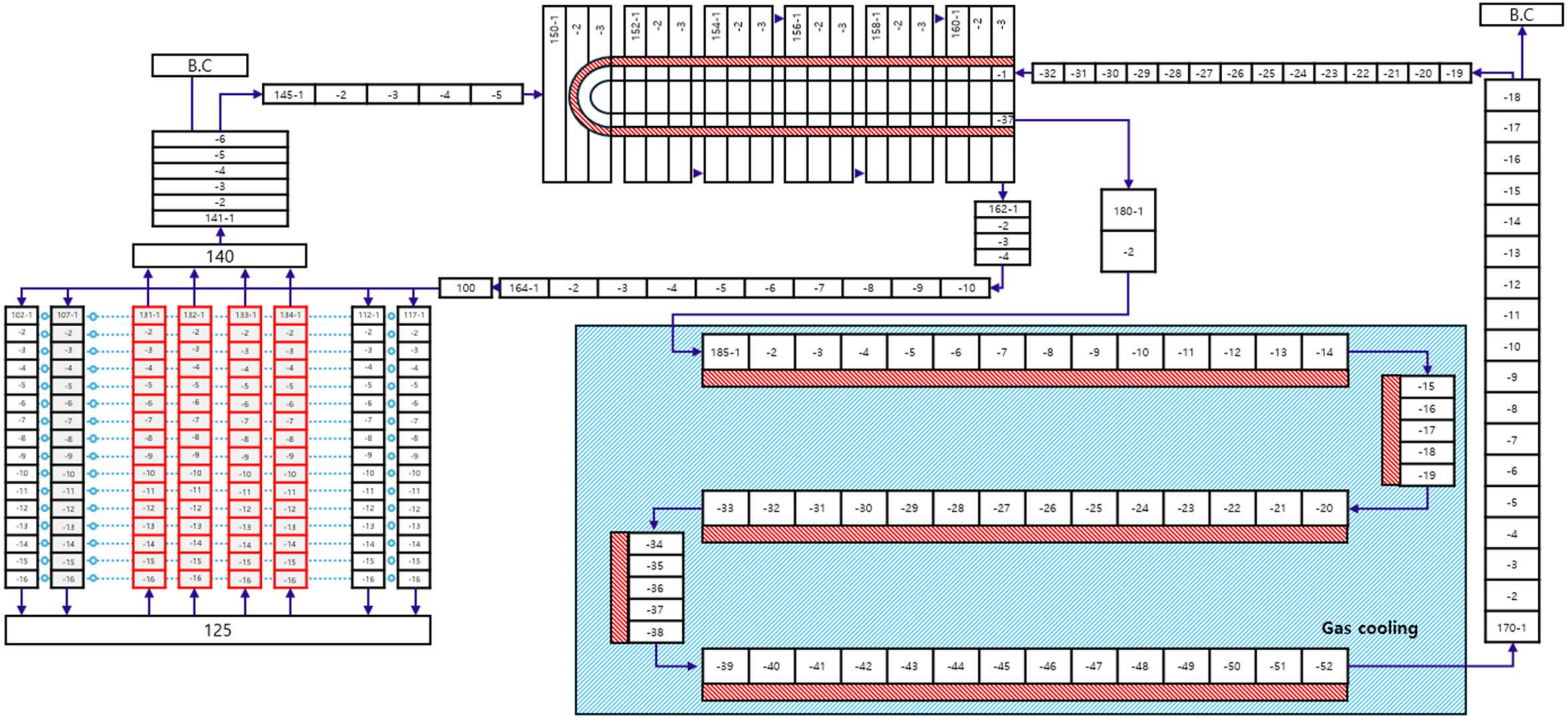


Fig. Schematic of MSRE nodalization for MARS-KS analysis

Configuration of analysis condition

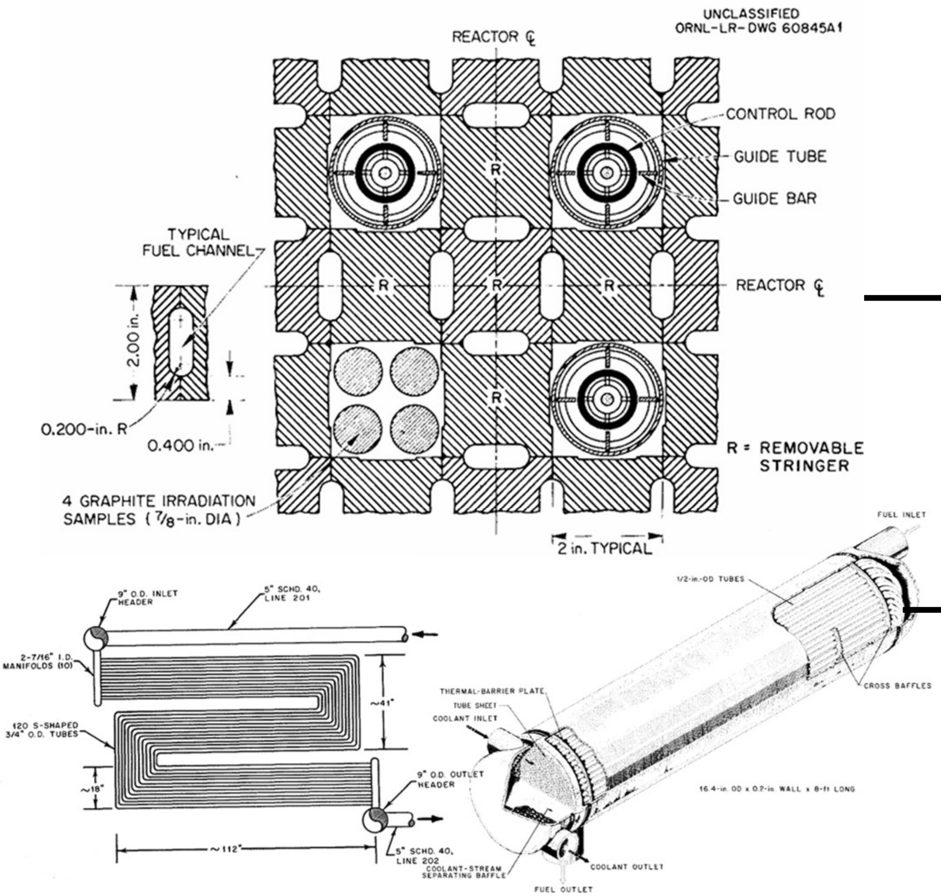


Fig. Schematics of MSRE components

Table. Summary of MSRE configuration

Operating parameter	Value	Unit
Core power	10	MW
Primary side pressure	2.413×10^5	Pa
Graphite channel hydraulic diameter	15.85	mm
Graphite channel height	1.73	m
Primary side flowrate	1200	gpm
Heat generation rate [Graphite]	6	%
Heat generation rate [Fuel]	94	%
Heat exchanger baffle distance	0.3048	m
Heat exchanger tube diameter	12.7	mm
Heat exchanger pressure	3.792×10^5	Pa
Heat exchanger flowrate	850	gpm
Effective heat transfer area of heat exchanger	24.06	m^2
Radiator tube diameter	19.05	mm
Radiator tube flowrate	200,000	cfm
Effective heat transfer area of radiator	65.59	m^2

Configuration of analysis condition

- In this study, the feasibility of MSR module analysis was demonstrated using MARS-KS, compared with the experimental data.

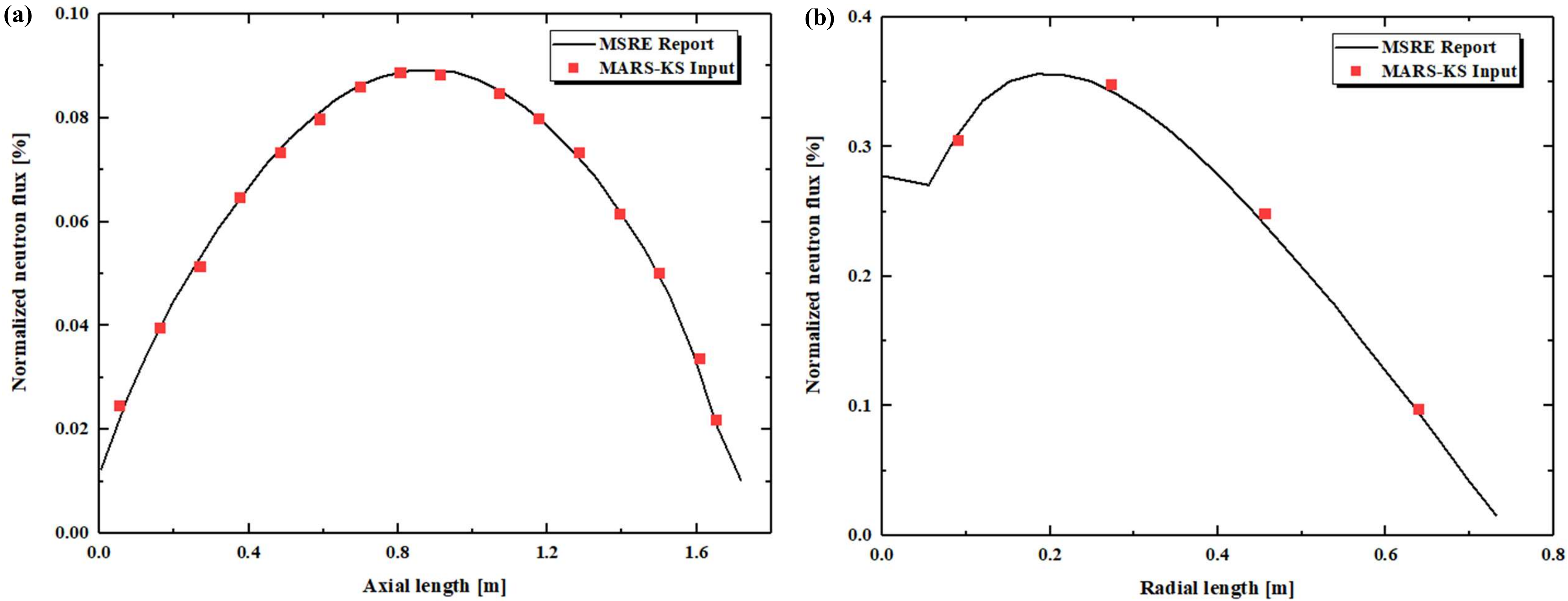


Fig. Normalized neutron flux distribution of MSRE and MARS-KS simulation: (a) axial and (b) radial

Modeling and Methodology

Configuration of Module analysis condition

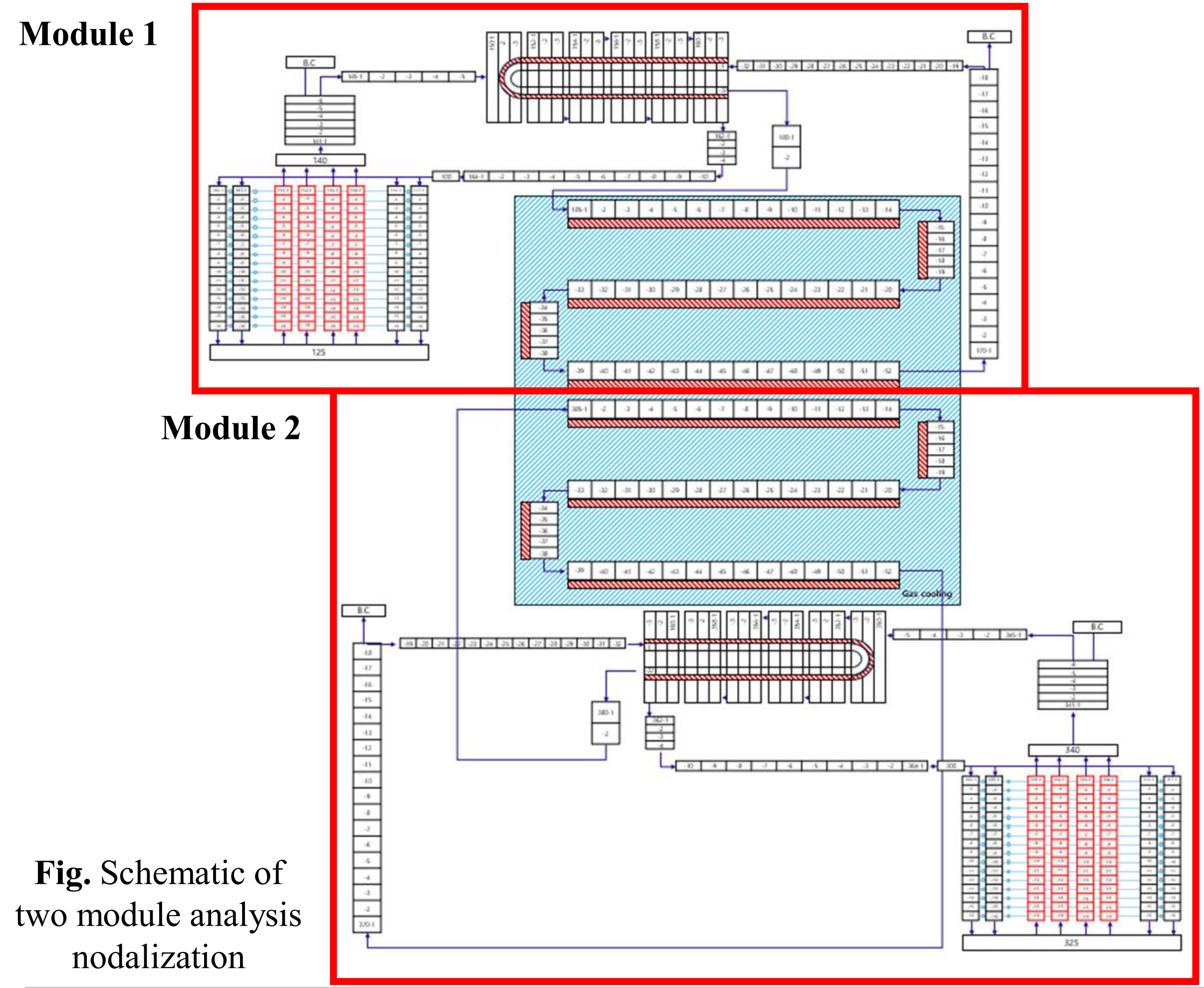


Fig. Schematic of two module analysis nodalization

Table. Summary of two modules analysis configuration

Operating parameter	Value	Unit
Core power	10	MW
Primary side pressure	2.413×10^5	Pa
Graphite channel hydraulic diameter	15.85	mm
Graphite channel height	1.73	m
Primary side flowrate	1200	gpm
Heat generation rate [Graphite]	6	%
Heat generation rate [Fuel]	94	%
Heat exchanger baffle distance	0.3048	m
Heat exchanger tube diameter	12.7	mm
Heat exchanger pressure	3.792×10^5	Pa
Heat exchanger flowrate	850	gpm
Effective heat transfer area of heat exchanger	24.06	m^2
Radiator tube diameter	19.05	mm
Radiator tube flowrate	**400,000**	**cfm**
Effective heat transfer area of radiator	**131.18**	$\boldsymbol{m^2}$

Single MSRE analysis results

Table. Comparison of heat transfer coefficient equation

Operating parameter	MSRE	Dittus Bolter Equation	Error [%]	Gnielinski Correlation	Error [%]
Core inlet temperature	908.15 K	971.48 K	6.97	911.77 K	0.40
Core outlet temperature	935.93 K	1002.6 K	7.12	937.55 K	0.17
Primary side pressure	2.413×10^5 Pa	2.421×10^5 Pa	0.33	2.426×10^5 Pa	0.54
Heat exchanger inlet temperature	824.82 K	833.87 K	1.09	823.29 K	0.19
Heat exchanger outlet temperature	866.48 K	869.85 K	0.39	859.33 K	0.83
Heat exchanger pressure	3.792×10^5 Pa	3.795×10^5 Pa	0.08	3.795×10^5 Pa	0.08

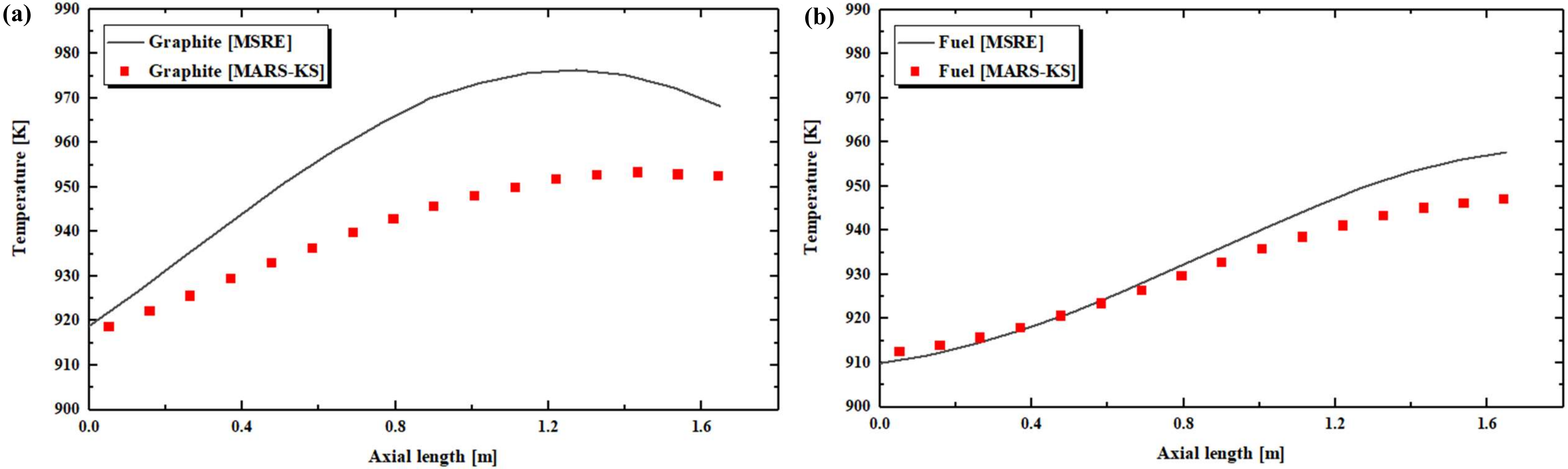


Fig. Comparison of axial temperature distributions between MSRE and MARS-KS predictions: (a) graphite moderator and (b) fuel salt

MSRE two modules analysis result at steady state condition

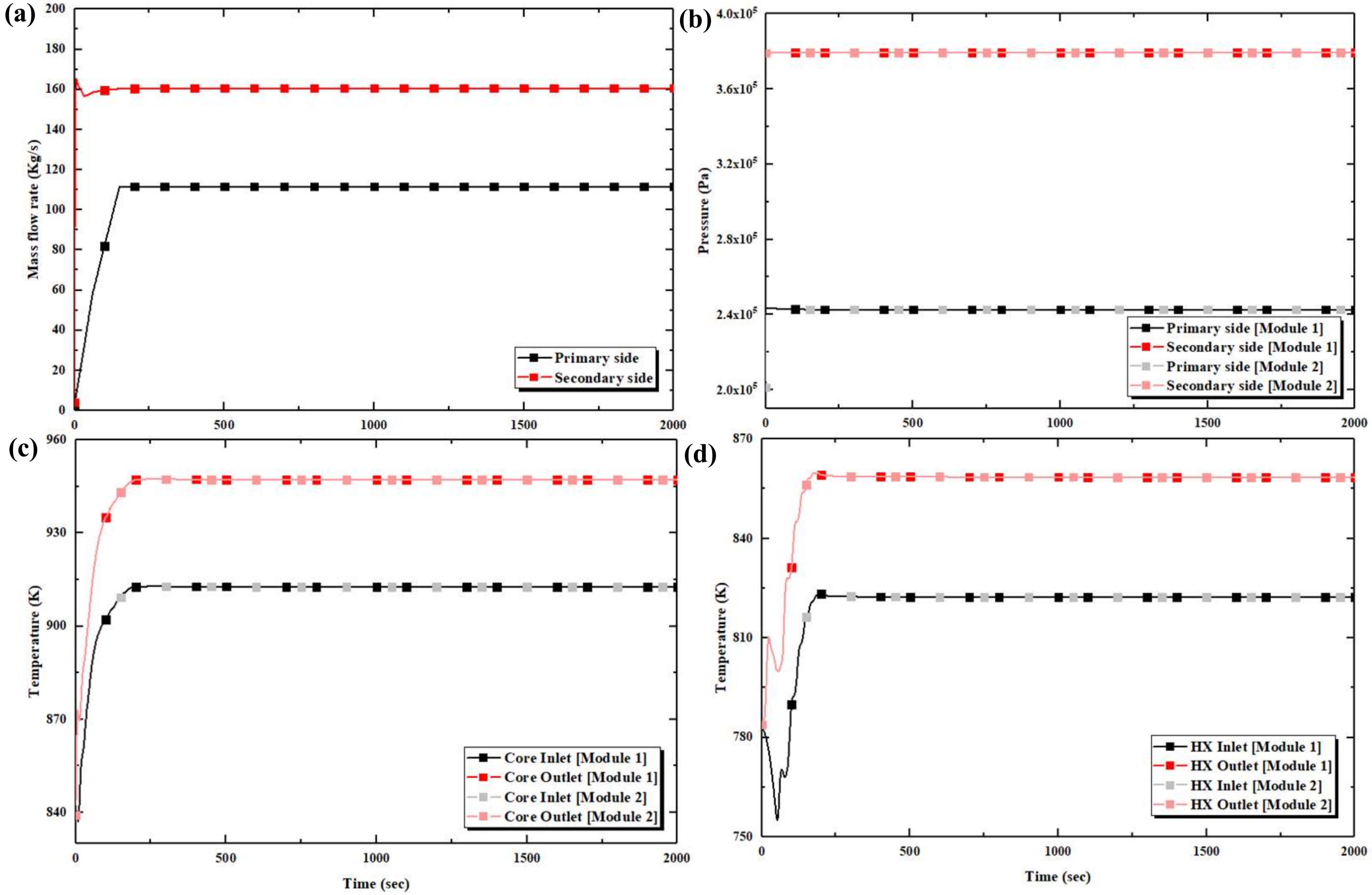


Table. Comparison with MSRE report

Operating parameter	MSRE	1 & 2	Error [%]
RV inlet temperature	908.15 K	911.77 K	0.40
RV outlet temperature	935.93 K	937.55 K	0.17
Primary side pressure	241.3 kPa	242.6 kPa	0.54
HX inlet temperature	824.82 K	823.29 K	0.19
HX outlet temperature	866.48 K	859.33 K	0.83
HX pressure	379.2 kPa	379.5 kPa	0.08

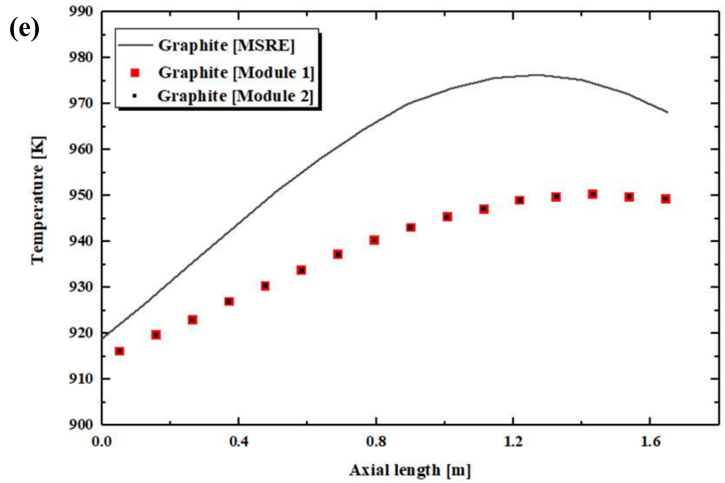


Fig. Behavior of main parameters until steady state condition: (a) mass flow rate, (b) pressure, (c) core temp., (d) HX temp., and (e) graphite temp.

3 Results and discussions

Feasibility of FLiNaK analysis

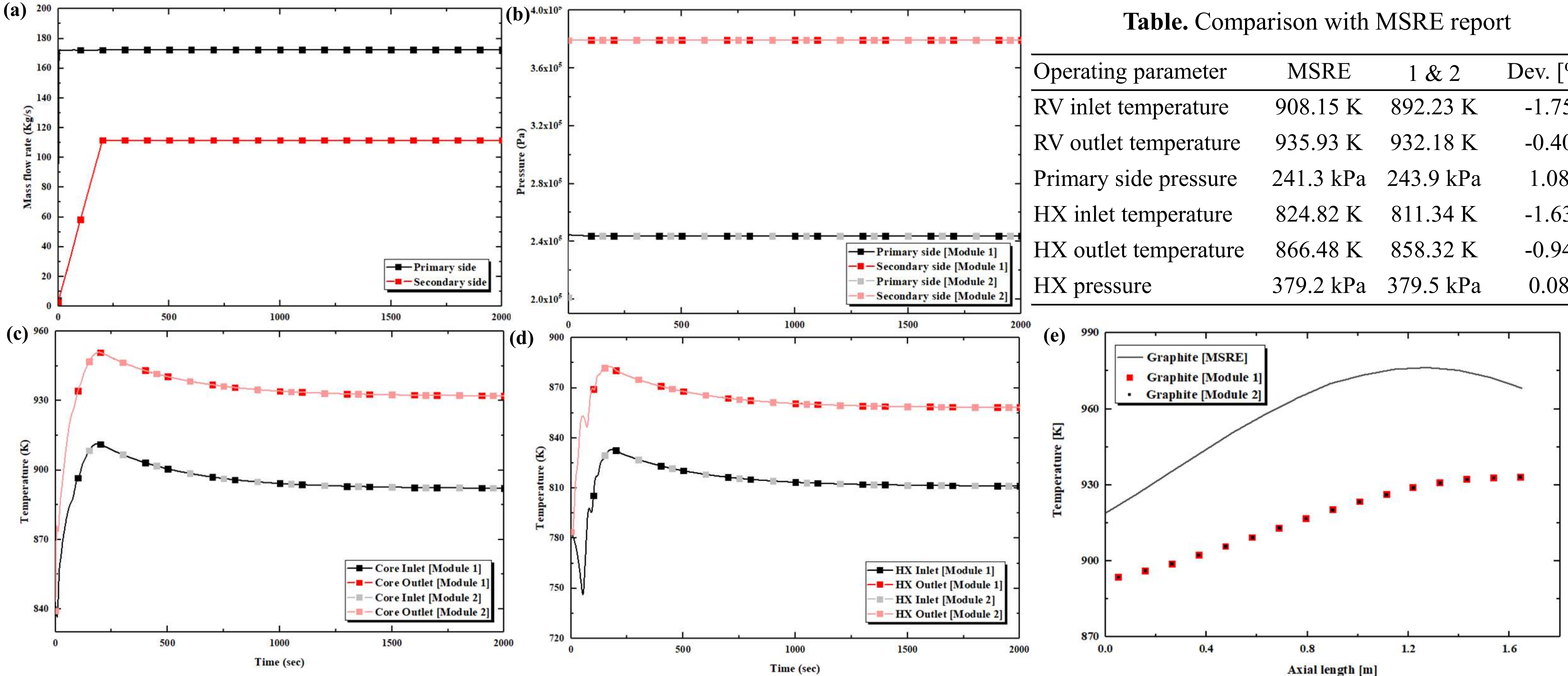


Table. Comparison with MSRE report

Operating parameter	MSRE	1 & 2	Dev. [%]
RV inlet temperature	908.15 K	892.23 K	-1.75
RV outlet temperature	935.93 K	932.18 K	-0.40
Primary side pressure	241.3 kPa	243.9 kPa	1.08
HX inlet temperature	824.82 K	811.34 K	-1.63
HX outlet temperature	866.48 K	858.32 K	-0.94
HX pressure	379.2 kPa	379.5 kPa	0.08

Fig. Behavior of main parameters until steady state condition: (a) mass flow rate, (b) pressure, (c) core temp., (d) HX temp., and (e) graphite temp.

Results and discussions

Feasibility of NaCl-KCl-$MgCl_2$ analysis

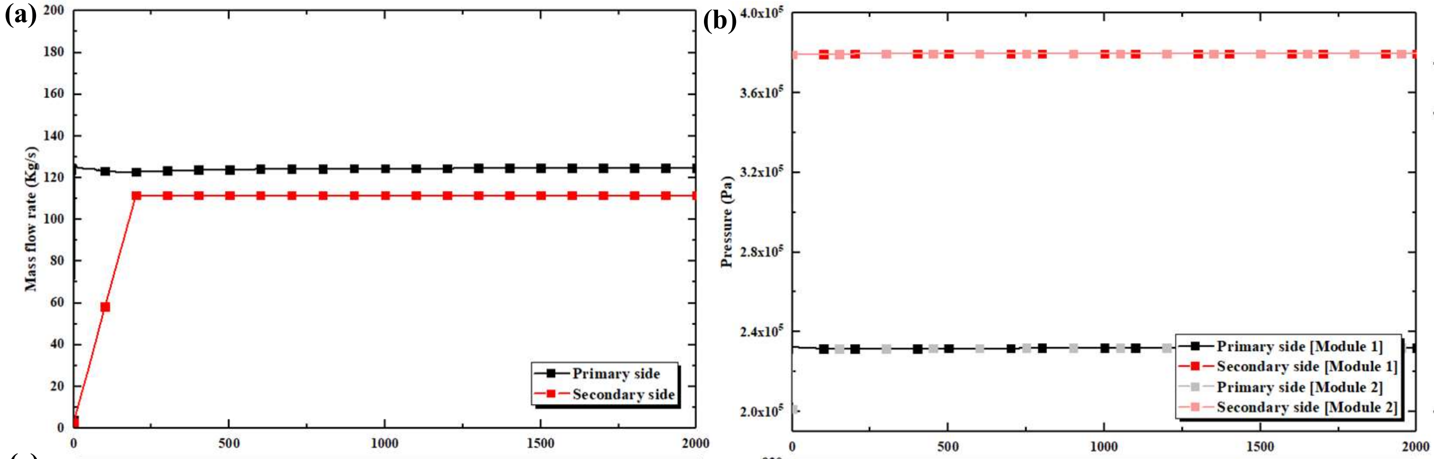


Table. Comparison with MSRE report

Operating parameter	MSRE	1 & 2	Dev. [%]
RV inlet temperature	908.15 K	883.01 K	-2.77
RV outlet temperature	935.93 K	943.69 K	0.83
Primary side pressure	241.3 kPa	231.9 kPa	3.90
HX inlet temperature	824.82 K	804.23 K	-2.50
HX outlet temperature	866.48 K	855.28 K	-1.29
HX pressure	379.2 kPa	379.5 kPa	0.08

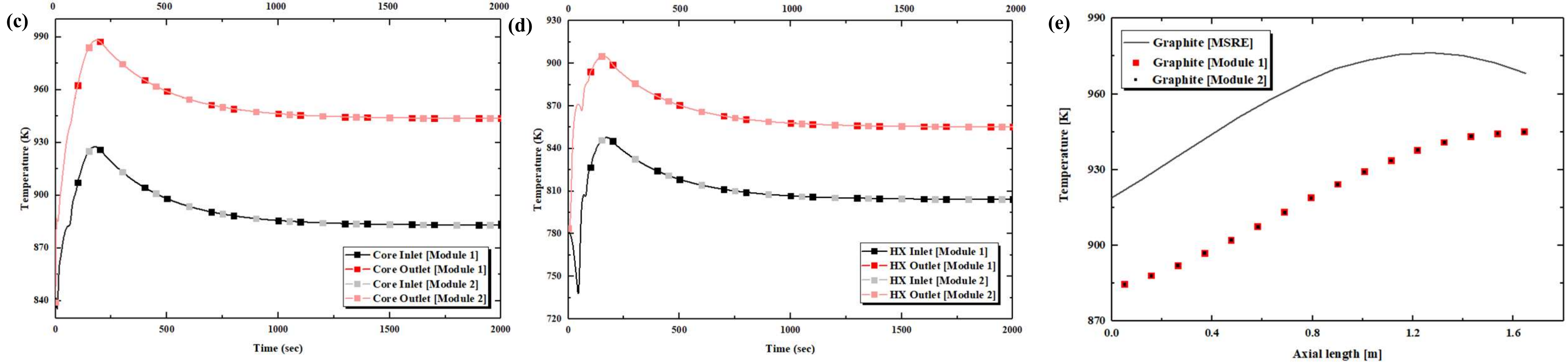


Fig. Behavior of main parameters until steady state condition: (a) mass flow rate, (b) pressure, (c) core temp., (d) HX temp., and (e) graphite temp.

Summary and future works

Summary

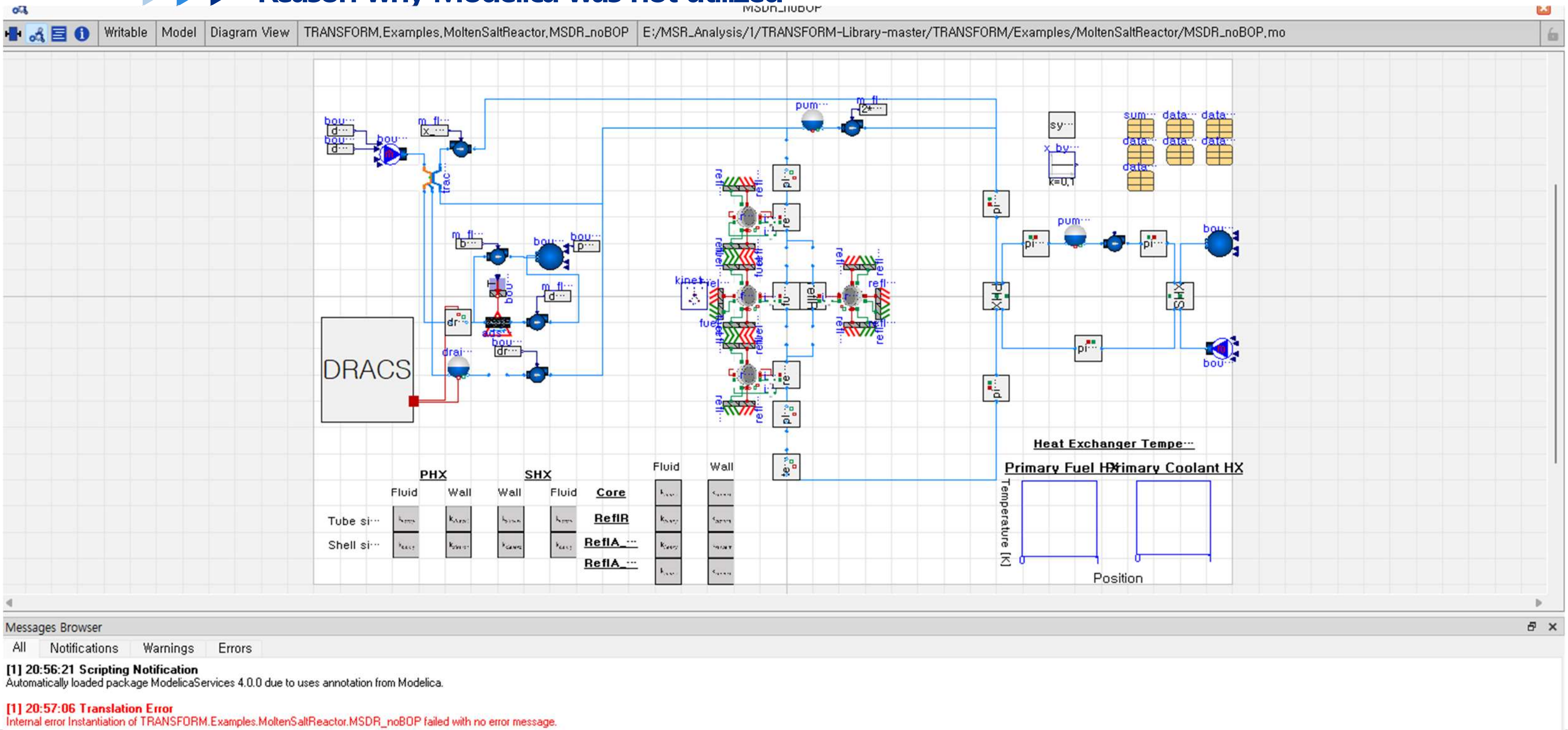
- The thermophysical properties of FLiBe, FLiNaK, and NaCl–KCl–$MgCl_2$ were implemented into MARS-KS, enabling steady-state analysis of molten salt reactor systems using multiple molten salt coolants.
- Steady state analysis was conducted for single and two module MSRE configurations, and the predicted thermal hydraulic parameters showed good agreement with the reference MSRE operating conditions.
- The results demonstrate the feasibility of utilizing MARS-KS for MSR module analysis and provide a basis for future investigations of more complex MSR systems.

Future works

- Transient analyses will be performed to investigate the influence of improved reactivity feedback models on reactor power, temperature, and flow behavior under representative operating scenarios.
- The current implementations of Doppler and density reactivity feedback will be refined to improve the physical consistency of the reactor kinetics model.
- The proposed analysis framework will be further validated under transient conditions and extended to evaluate the behavior and interaction of multiple MSR modules.

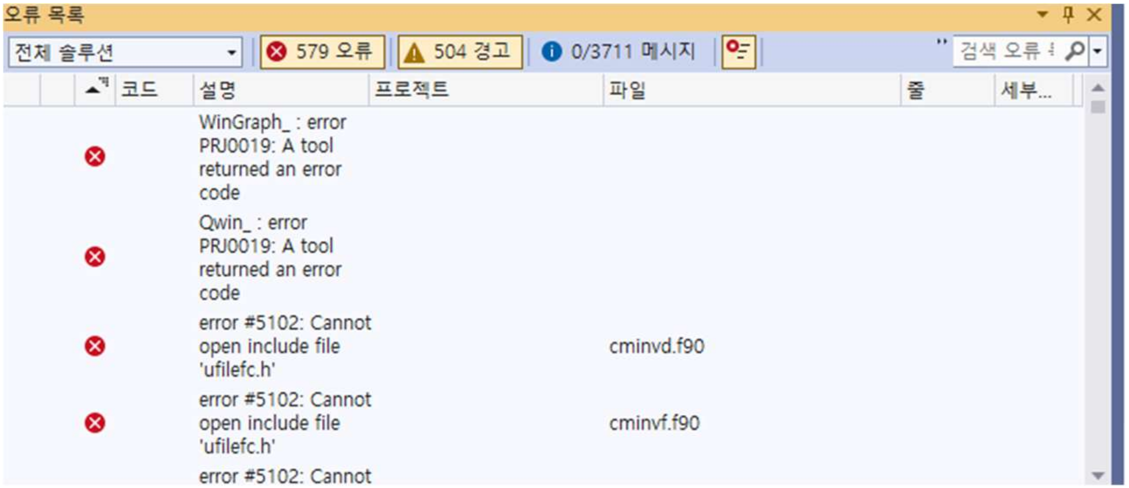
Thanks for your attention.

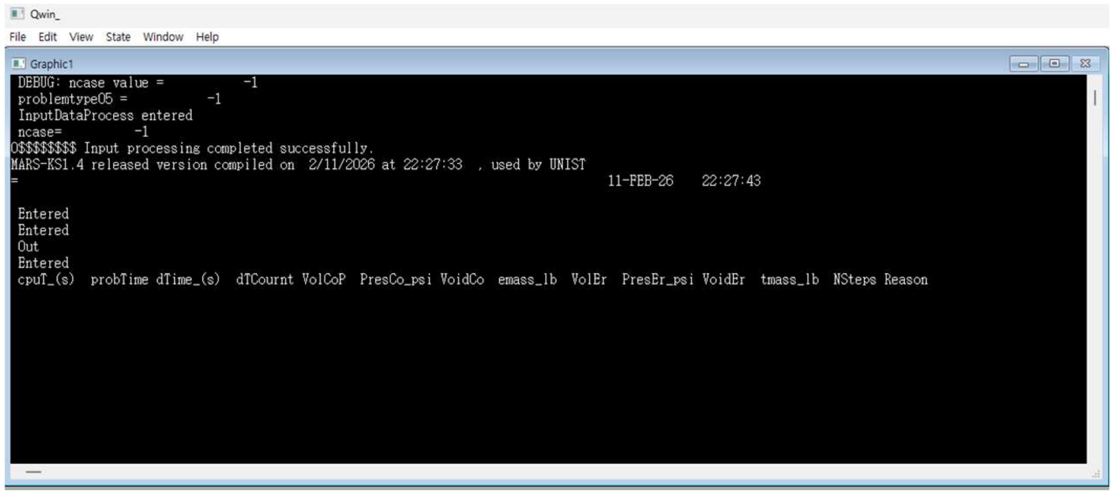
Reason why Modelica was not utilized



Appendix

Process for MARS-KS source code modification





Fortran for Win32 is no longer being supplied, so source code should be modified to process win 64.

확인

1. **MARS-KS was designed for Win 32 bit. So, the window operator was changed for hundreds of files**
2. **The parameter and method to read input file was modified to work at win 64 bit environment for hundreds of files**
3. **The plot visualization and save output & plotfl files don't work at win 64 bit. So, visualization function and save function was changed for hundreds of files.**